DELIVERY LOGISTICS & PERFORMANCE

DATA ANALYTICS REPORT

Dataset profiling

- Data quality
- Exploratory analysis
- Business recommendations

KEY METRIC	VALUE	KEY METRIC	VALUE
Records analysed	9,999	Columns	13
Missing values	0	Exact duplicate rows	0
Average delivery cost	809.32	Average distance	141.32 km

1. Executive Summary

This report analyses 9,999 delivery records across 13 fields in Delivery_Logistics_Analysis.csv. The file contains 0 missing cells and 0 exact duplicate rows. The average delivery cost is 809.32, median cost is 789.65, average distance is 141.32 km, and average customer rating is 4.74 out of 5.

The strongest numerical relationship is between delivery distance and delivery cost (Pearson correlation 0.990), indicating that longer journeys generally have higher recorded costs. This is an association, not proof that distance is the only cost driver.

A key limitation is that the dataset records only delivered (9,999) status outcomes. If the data contains only delivered records, it cannot be used to estimate a real-world delay rate or failure rate. The customer ratings also cover only the values observed in this file.

Main findings

- Delivery records: 9,999; fields: 13.
- Average delivery cost: 809.32; median: 789.65.
- Average distance: 141.32 km; average package weight: 25.07 kg.
- Average customer rating: 4.74/5; observed ratings: 4, 5.
- Distance-cost correlation: 0.990, a very strong positive association.
- 1,175 records are flagged by the IQR rule for cost per km; investigate rather than automatically delete them.

Project objective

To profile the delivery dataset, evaluate data quality, identify cost and volume patterns, compare categories, and provide evidence-based recommendations for a Power BI dashboard and further operational analysis.

2. Dataset Overview

The dataset describes deliveries using an identifier, delivery partner, package and vehicle categories, delivery mode, region, weather condition, distance, package weight, status, customer rating, delivery cost, and cost per kilometre.

Column	Field type	Business meaning / analytical use
delivery_id	Identifier	Record-level reference; not a measure to average.
delivery_partner	Categorical	Compare delivery volume, average cost, and rating.
package_type	Categorical	Compare package mix, weight, and cost.
vehicle_type	Categorical	Compare vehicle usage and cost patterns.
delivery_mode	Categorical	Compare standard, two-day, same-day, and express services.
region	Categorical	Compare volume and cost by geography.
weather_condition	Categorical	Explore observed cost differences by weather.
distance_km	Numeric	Delivery distance in kilometres.
package_weight_kg	Numeric	Package weight in kilograms.
delivery_status	Categorical	Delivery outcome; limited if only one status exists.
delivery_rating	Numeric / ordinal	Customer rating; interpret as an ordered score.
delivery_cost	Numeric	Recorded delivery cost/charge; currency not specified.
cost_per_km	Numeric	Recorded cost per kilometre; validate against cost ÷ distance.

Data types

Treat delivery_id as text even though it looks like an identifier. Use categorical fields as dimensions/slicers and numerical fields as measures. Ratings are ordinal and should be interpreted carefully.

3. Data Quality Assessment

Check	Result	Interpretation / action
Row and column count	9,999 rows × 13 columns	Dataset size confirmed from the CSV.
Missing cells	0	No imputation is needed for missing values if this result remains unchanged after refresh.
Exact duplicate rows	0	No full-row duplicates detected.
Unique delivery IDs	9,999 of 9,999	Check that IDs are unique before using DISTINCTCOUNT as the main record count.
Observed status values	delivered: 9,999 (100.0%)	Status diversity determines whether delay/failure KPIs are possible.
Observed rating values	4, 5	Only observed values should be reported; check whether lower scores are missing by design.
Cost-per-km IQR flags	1,175 (fences 3.52 to 8.43)	Statistical flags are candidates for review, not automatic errors.
Cost/km calculation check	100.00% within 0.02 of cost ÷ distance	Small differences may be due to rounding or source rules.

Data preparation recommendations

- Trim leading/trailing spaces and standardise category spelling/capitalisation in Power Query.
- Set delivery_id to Text; set distance, weight, cost, and cost-per-km to Decimal Number; set rating to Whole Number.
- Check for zero or negative distance before calculating cost per kilometre.
- Confirm the business definition and currency of delivery_cost.
- Investigate extreme cost-per-km records against minimum charges, route fees, or source-data issues.
- Do not remove records solely because they are statistical outliers; document any business-approved exclusion.

4. Numerical Summary

Metric	Mean	Median	Minimum	Maximum	Std. dev.
distance_km	141.32	136.60	3.60	297.10	85.31
package_weight_kg	25.07	25.00	0.67	49.52	14.42
delivery_rating	4.74	5.00	4.00	5.00	0.44
delivery_cost	809.32	789.65	95.67	1,632.72	423.77
cost_per_km	7.17	5.71	5.00	73.49	5.34

Average delivery cost (809.32) is above the median (789.65), suggesting a right-skewed distribution or a subset of high-cost deliveries. Average cost per kilometre is 7.17, while its median is 5.71; this gap is consistent with high-ratio records

influencing the mean.



Figure 1. Each point is a delivery record; the line shows the fitted linear trend.

The distance-cost correlation is 0.990. The correlation between distance and cost per kilometre is -0.446, and between package weight and delivery cost is 0.090. These correlations describe association only and do not establish causality.

5. Delivery Partner Analysis

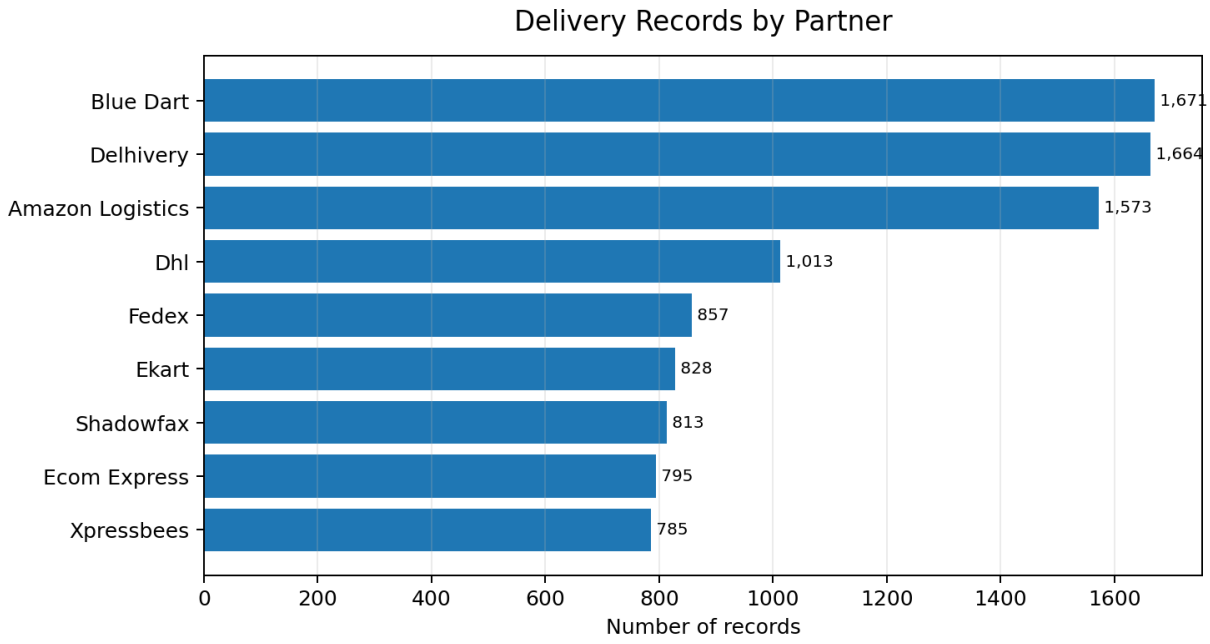


Figure 2. Delivery record counts by partner.

Partner	Records	Avg. cost	Avg. rating	Avg. distance (km)
Blue Dart	1,671	794.93	4.51	138.14
Delhivery	1,664	801.39	4.50	139.69
Amazon Logistics	1,573	817.54	4.49	143.23
Dhl	1,013	811.33	4.83	141.51
Fedex	857	824.57	5.00	144.38
Ekart	828	844.45	5.00	148.22
Shadowfax	813	807.50	5.00	141.13
Ecom Express	795	804.50	5.00	140.54
Xpressbees	785	790.71	5.00	137.77

Blue Dart has the highest record count (1,671), while Xpressbees has the lowest (785). Ekart has the highest average cost and Xpressbees the lowest. These comparisons should be interpreted alongside route distance, package mix, vehicle type, and service mode.

6. Delivery Mode Analysis

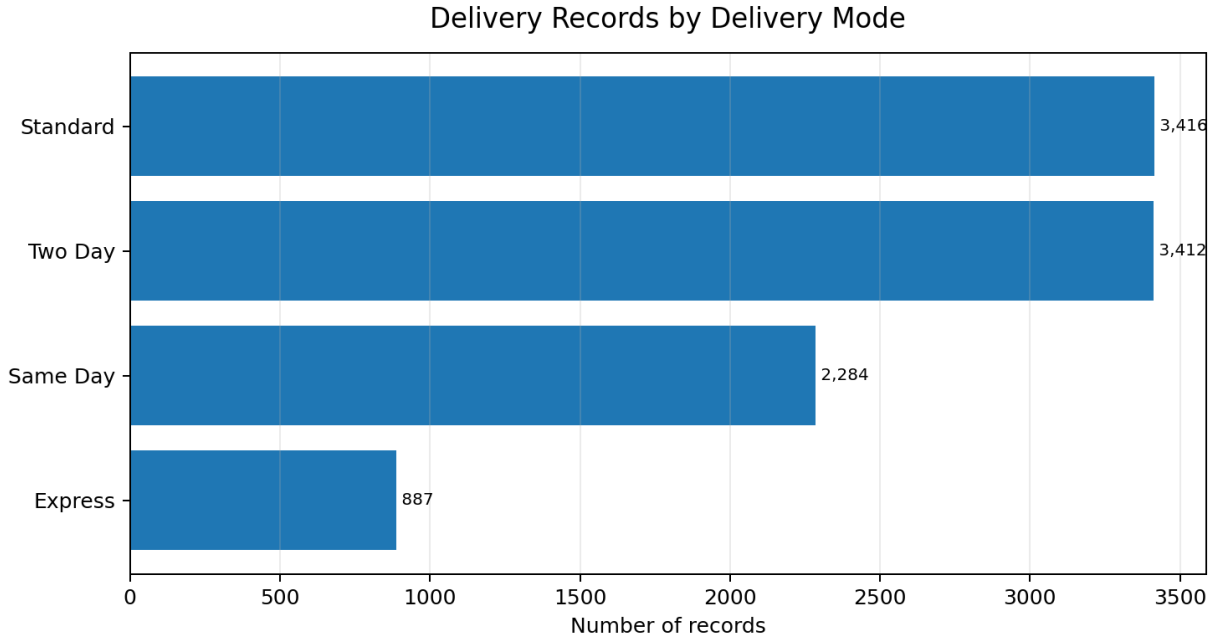


Figure 3. Record counts by delivery mode.

Mode	Records	Share	Avg. cost	Avg. cost/km	Avg. distance (km)
Standard	3,416	34.16%	816.59	6.41	148.06
Two Day	3,412	34.12%	825.12	6.41	149.99
Same Day	2,284	22.84%	772.84	9.30	119.57
Express	887	8.87%	814.46	7.48	137.96

Standard and two-day are the most frequently represented modes in this file. Compare average cost and cost per kilometre, but account for distance and weight before interpreting the differences as service efficiency.

7. Regional and Weather Analysis

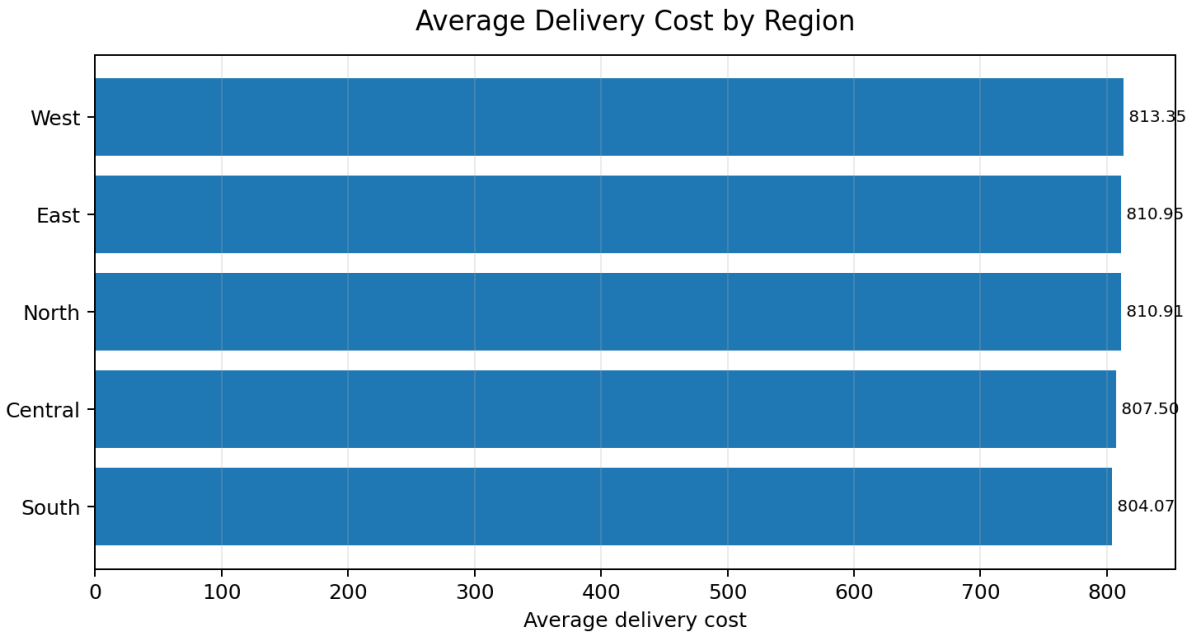


Figure 4. Average delivery cost by region.

Region	Records	Average cost	Average distance (km)
Central	2,185	807.50	140.61
East	1,954	810.95	141.64
North	1,951	810.91	141.86
South	1,952	804.07	140.49
West	1,957	813.35	142.06

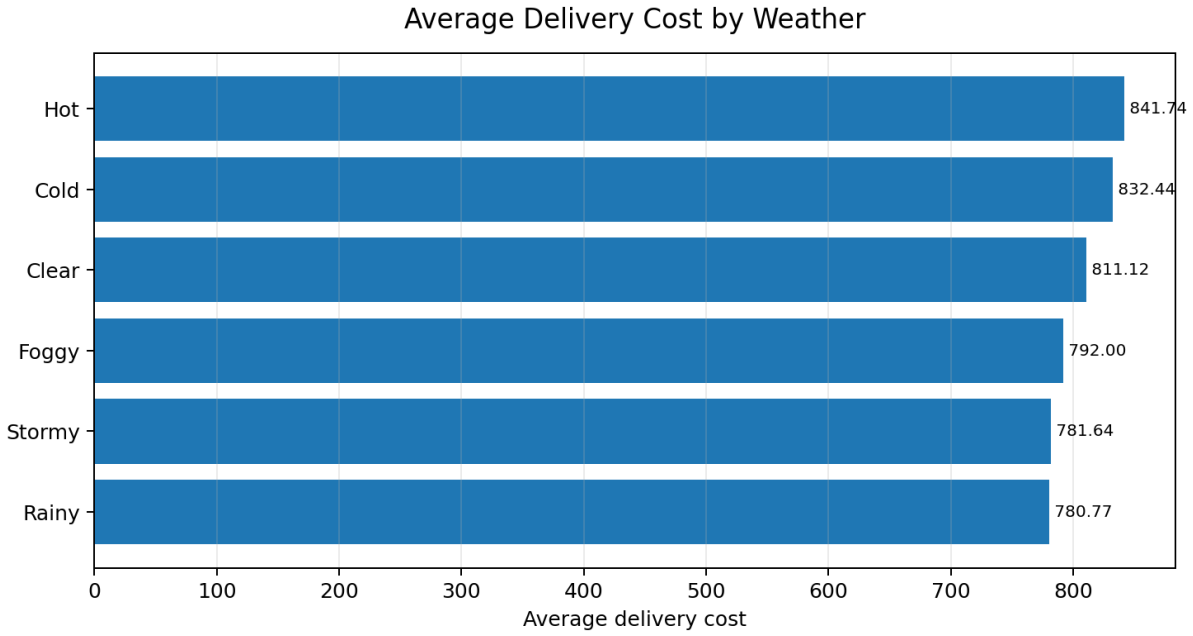


Figure 5. Average delivery cost by weather condition.

Weather	Records	Average cost	Average cost/km
Hot	1,861	841.74	7.15
Cold	1,877	832.44	7.05
Clear	1,828	811.12	7.46

Foggy	1,631	792.00	7.16
Stormy	1,338	781.64	7.06
Rainy	1,464	780.77	7.06

Regional and weather comparisons are descriptive. Different average distances, package mixes, and service modes can explain some observed differences; the dataset does not establish that region or weather caused a cost change.

8. Package and Vehicle Analysis

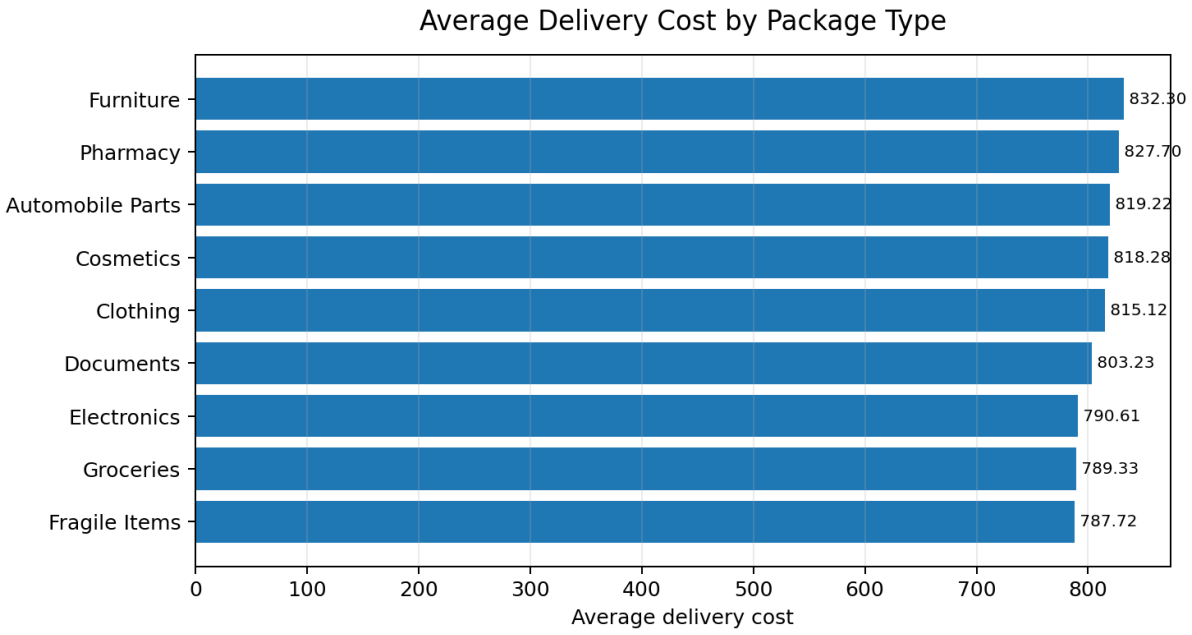


Figure 6. Average delivery cost by package type.

Package type	Records	Average cost	Average weight (kg)
Clothing	1,144	815.12	25.12
Automobile Parts	1,136	819.22	25.02
Fragile Items	1,136	787.72	25.03
Documents	1,129	803.23	24.85
Pharmacy	1,124	827.70	24.75
Electronics	1,111	790.61	24.90
Furniture	1,095	832.30	25.09
Cosmetics	1,068	818.28	25.85
Groceries	1,056	789.33	25.04

Vehicle type	Records	Average cost	Average distance (km)
Truck	1,693	816.44	142.89
Ev Van	1,682	797.24	138.82
Scooter	1,679	816.44	142.68
Ev Bike	1,673	804.47	139.94
Van	1,641	801.56	139.83
Bike	1,631	819.82	143.76

Package categories and vehicle types show differences in average cost. To make fair comparisons, segment by distance bands, delivery mode, and package weight. Recorded delivery cost is not necessarily the same as the logistics provider's operating cost.

9. Delivery Status and Customer Ratings

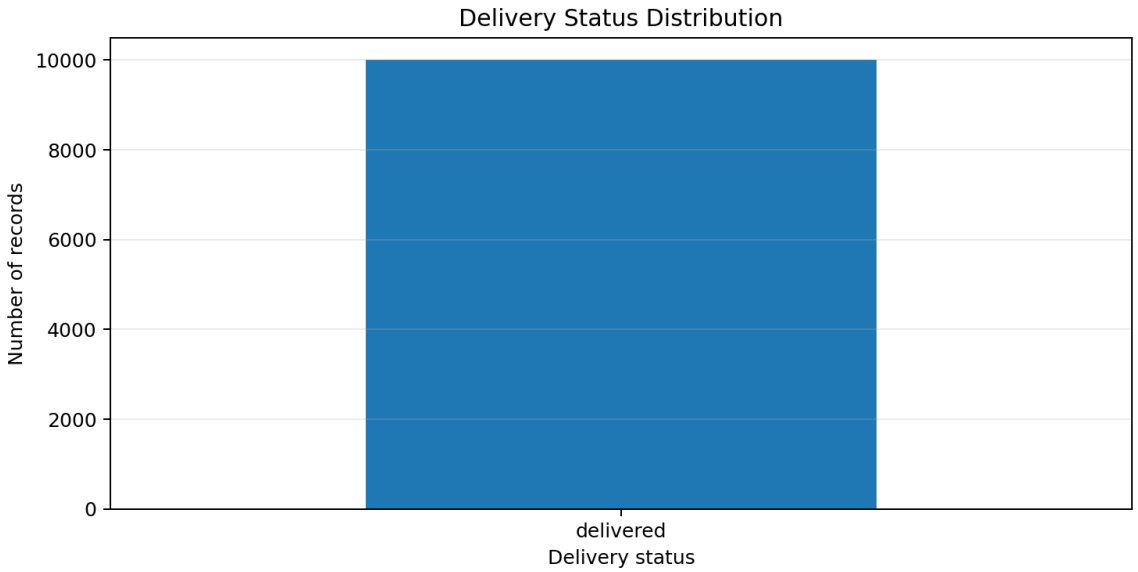


Figure 7. Status distribution in the uploaded file.

Status	Records	Share
Delivered	9,999	100.00%

Average customer rating is 4.74/5. The observed rating range is 4–5. If the file was filtered to include only successful deliveries or high ratings, these results would overstate overall satisfaction and should not be generalised to all deliveries.

The current file does not provide enough evidence to estimate delivery delay or failure rates unless the status field contains those outcomes. For a robust SLA analysis, add promised delivery timestamp, actual delivery timestamp, cancellation/failure reason, and a complete sample of delivery outcomes.

10. Business Recommendations

1. **Validate pricing rules:** Review high cost-per-kilometre records, especially short-distance deliveries, and confirm whether minimum charges or surcharges apply.
2. **Compare like with like:** Benchmark partners and vehicles within comparable distance, weight, region, and delivery-mode groups.
3. **Build operational KPIs:** After obtaining complete status and timestamp data, calculate on-time delivery %, delay rate, failed delivery %, and average delay hours.
4. **Improve rating coverage:** Confirm why only a limited rating range appears; collect ratings from all eligible customers and track response rates.
5. **Track cost efficiency:** Use total delivery cost ÷ total distance as a weighted cost-per-kilometre KPI, alongside the average of row-level ratios.
6. **Add time-based analysis:** Capture order date, dispatch time, and delivery completion time to analyse daily/weekly trends and peak demand.
7. **Document data definitions:** Record currency, cost definition, rating scale, delivery-mode definitions, and data extraction filters.

11. Recommended Power BI Dashboard

Suggested one-page layout:

- Top KPI cards: total deliveries, average delivery cost, average distance, average rating, weighted cost per km.
- Partner performance: horizontal bar chart for delivery volume and a separate chart for average cost.
- Cost drivers: scatter plot of distance vs delivery cost; optional trend line.
- Delivery mix: delivery mode, package type, and vehicle type charts.
- Geographic and context view: average cost by region and weather condition.
- Slicers: delivery partner, region, delivery mode, vehicle type, package type, and weather condition.

Suggested DAX measures

Total Deliveries

```
Total Deliveries = DISTINCTCOUNT(Delivery_Logistics_Analysis[delivery_id])
```

Average Delivery Cost

```
Average Delivery Cost = AVERAGE(Delivery_Logistics_Analysis[delivery_cost])
```

Average Distance

```
Average Distance = AVERAGE(Delivery_Logistics_Analysis[distance_km])
```

Average Rating

```
Average Rating = AVERAGE(Delivery_Logistics_Analysis[delivery_rating])
```

Weighted Cost per KM

```
Weighted Cost per KM = DIVIDE(SUM(Delivery_Logistics_Analysis[delivery_cost]),
SUM(Delivery_Logistics_Analysis[distance_km]))
```

Replace the table name with the exact name in the Power BI Fields pane if it differs.

12. Conclusion

The analysis of 9,999 delivery records across 13 columns provides a useful starting point for logistics cost and delivery-mix analysis. The strongest finding is the positive distance-cost relationship (r = 0.990). The data also supports comparisons by delivery partner, mode, region, weather, package category, and vehicle type.

Data quality checks found 0 missing cells and 0 exact duplicate rows. However, the 1,175 IQR-flagged cost-per-kilometre records and the restricted observed rating/status values should be reviewed before using this report for operational decisions.

The recommended next step is to validate the business rules, then build the Power BI dashboard. For an interview-ready project, extend the dataset with actual and promised delivery times, all delivery outcomes, and richer customer feedback. This will support a more complete view of cost efficiency, reliability, and customer experience.

Appendix A. Interpretation Notes

- Counts represent rows in the uploaded CSV unless otherwise stated.
- Average costs are shown in the dataset's numeric units; currency was not supplied.
- Correlations measure linear association and do not establish cause and effect.
- IQR outliers are flagged for investigation, not automatically classified as data errors.
- Dashboard measures should be refreshed and revalidated after any cleaning or filtering.
- This report is based on the uploaded CSV file; it does not claim to verify the contents or execution state of a separate Colab notebook.